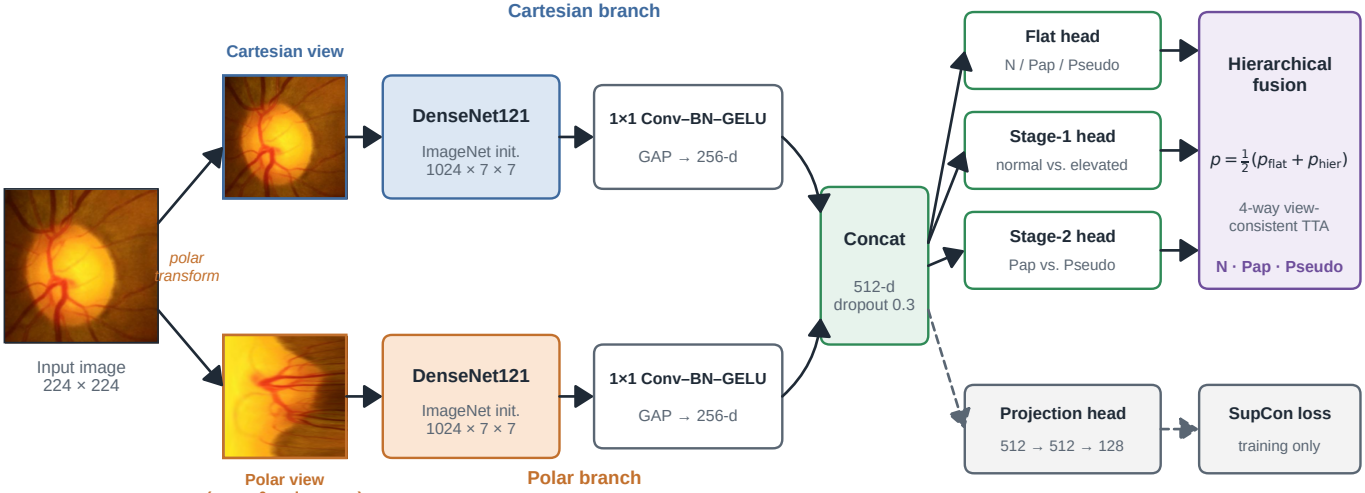


Training loss: $\mathcal{L} = \mathrm{CE}_w(\text{flat}, y) + \lambda_h\,[\mathrm{CE}(\text{stage-1}) + \mathrm{CE}(\text{stage-2} \mid \text{elevated})] + \lambda_s\,\mathcal{L}_{\text{SupCon}}, \quad \lambda_h = 0.5,\ \lambda_s = 0.3$